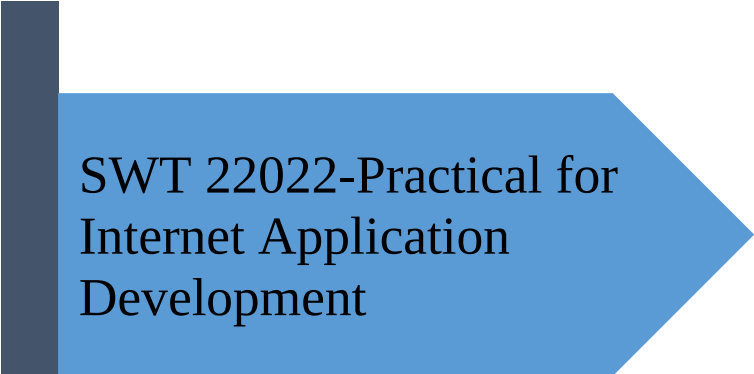




SWT 22022-Practical for
Internet Application
Development

Department of Information
and Communication
Technology
Faculty of Technology



Lab sheet :08
Reg. Number: SEU/IS/20/ICT/084
Academic Year :2020/2021
Date: 2024.07.08
Practical No :08

Title:

- PHP, MySQL and JavaScript.

Aims:

- Create and read data with PHP and MySQL
- Creating a simple web application using PHP, MySQL and JavaScript

Tasks:

- Create new database
- Implement functions for store data
- Implement functions for read specific user data
- Save sessions when user login

Activities :

1. Create a new database as “**expense_tracker**”
2. Create a new table named “**users**”

```
CREATE DATABASE expense_tracker;
USE expense_tracker;
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    password VARCHAR(255) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

localhost / 127.0.0.1 | phpMyAdmin x Students

localhost/phpmyadmin/index.php?route=/server/sql

Server: 127.0.0.1

Run SQL query/queries on server "127.0.0.1":

```
1 CREATE DATABASE expense_tracker;
2 USE expense_tracker;
3 CREATE TABLE users (
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     username VARCHAR(50) NOT NULL UNIQUE,
6     password VARCHAR(255) NOT NULL,
7     email VARCHAR(100) NOT NULL UNIQUE,
8     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
9 );
10
```

Clear Format Get auto-saved query

☐ Bind parameters

Console this SQL query:

localhost / 127.0.0.1 / expense_ x Students

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=expense_tracker&table=users

Server: 127.0.0.1 » Database: expense_tracker » Table: users

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0010 seconds.)

`SELECT * FROM `users``

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

id	username	password	email	created_at
----	----------	----------	-------	------------

Query results operations

Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

3. Implement functions for user registration

I. JavaScript

```
async function register() {
  const username = document.getElementById('registerUsername').value;
  const password = document.getElementById('registerPassword').value;
  const email = document.getElementById('registerEmail').value;
  const response = await fetch('php/register.php', {
    method: 'POST',
    body: JSON.stringify({ username, password, email }),
    headers: { 'Content-Type': 'application/json' }
  });
  const result = await response.json();
  alert(result.message);
}
```

II. register.php

```
<?php
session_start();
require_once '../database/User.php';
$database = new Database("localhost", "root", "", "expense_tracker");
$db = $database->getConnection();
$user = new User($db);
$data = json_decode(file_get_contents("php://input"));
$user->setUsername($data->username); $user->setPassword($data->password); $user->setEmail($data->email);
if ($user->register()) {
  echo json_encode(["message" => "User was registered."]);
} else {
  echo json_encode(["message" => "Unable to register the user."]);
}
?>
```

4. Implement functions for user login

I. JavaScript

```
async function login() {
  const username = document.getElementById('loginUsername').value;
  const password = document.getElementById('loginPassword').value;
  const response = await fetch('php/login.php', {
    method: 'POST',
    body: JSON.stringify({ username, password }),
    headers: { 'Content-Type': 'application/json' }
  });
}
```

```

    });
    const result = await response.json();
    if (result.message === "Login successful.") {
        window.location.href = 'dashboard.html';
    } else {
        alert(result.message);
    }
}

```

II. login.php

```

<?php
    session_start();
    require_once '../database/User.php';
    $database = new Database("localhost", "root", "", "expense_tracker");
    $db = $database->getConnection();
    $user = new User($db);
    $data = json_decode(file_get_contents("php://input"));
    $user->setUsername($data->username); $user->setPassword($data->password);
    if ($user->login()) {
        $_SESSION['username'] = $user->username;
        echo json_encode(["message" => "Login successful."]);
    } else {
        echo json_encode(["message" => "Login failed."]);
    }
?>

```

5. Implement function for logout

I. JavaScript

```

async function logout() {
    const response = await fetch('php/logout.php', { method: 'POST' });
    const result = await response.json();
    if (result.message === "Logged out successfully.") {
        window.location.href = 'index.html';
    } else {
        alert(result.message);
    }
}

```

II. logout.php

```

<?php
    session_start();
    session_destroy();
    echo json_encode(["message" => "Logged out successfully."]);
?>

```

Code

Database.php

```
<?php
class Database {
    private $host;
    private $username;
    private $password;
    private $database;
    private $connection;

    public function __construct($host, $username, $password, $database) {
        $this->host = $host;
        $this->username = $username;
        $this->password = $password;
        $this->database = $database;

        $this->connect();
    }

    private function connect() {
        $this->connection = mysqli_connect($this->host, $this->username, $this->password,
$this->database);
        if (!$this->connection) {
            die("Server Error!");
        }
    }

    public function getConnection() {
        return $this->connection;
    }
}
?>
```

Login.php

```
<?php
session_start();
require_once '../database/Database.php';
require_once '../database/User.php';

$dbdatabase = new Database("localhost", "root", "", "expense_tracker");
$dbconnection = $dbdatabase->getConnection();
$user = new User($dbconnection);

$data = json_decode(file_get_contents("php://input"));
```

```

$user->setUsername($data->username);
$user->setPassword($data->password);

if ($user->login()) {
    $_SESSION['username'] = $user->username; // Corrected here
    echo json_encode(["message" => "Login success"]);
} else {
    echo json_encode(["message" => "Login fail"]);
}
?>

```

register.php

```

<?php
session_start();
require_once '../database/Database.php';
require_once '../database/User.php';

$databse = new Database("localhost", "root", "", "expense_tracker");
$db = $databse->getConnection();
$user = new User($db);

$data = json_decode(file_get_contents("php://input"));
$user->setUsername($data->username);
$user->setPassword($data->password);
$user->setEmail($data->email);

if ($user->register()) {
    echo json_encode(["message" => "User was registered."]);
} else {
    echo json_encode(["message" => "Unable to register the user."]);
}
?>

```

User.php

```

<?php
class User {
    public $username;
    public $password;
    public $email;
    public $id;

    private $conn;

```

```

public function __construct($db) {
    $this->conn = $db;
}

public function setUsername($username) {
    $this->username = $username;
}

public function setPassword($password) {
    $this->password = $password;
}

public function setEmail($email) {
    $this->email = $email;
}

public function register() {
    $hash_password = password_hash($this->password, PASSWORD_DEFAULT);
    $query = "INSERT INTO users (username, password, email) VALUES ('$this->username',
'$hash_password', '$this->email')";
    $result = mysqli_query($this->conn, $query);
    return $result !== false; // Return true on success
}

public function login() {
    $query = "SELECT id, username, password FROM users WHERE username = '$this->username'
LIMIT 1";
    $result = mysqli_query($this->conn, $query);

    if ($result) {
        $row = mysqli_fetch_assoc($result);
        if ($row && password_verify($this->password, $row['password'])) {
            $this->id = $row['id'];
            $this->username = $row['username'];
            return true;
        }
    }
    return false;
}
}
?>

```


Logout.php

```
<?php
session_start();
$_SESSION = [];
session_destroy();
header('Content-Type: application/json');
echo json_encode(["message" => "Logout success!"]);
?>
```

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Login and Registration</title>
  <!-- Bootstrap CSS -->
  <link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">

  <script>
    async function login() {
      const username = document.getElementById('loginUsername').value;
      const password = document.getElementById('loginPassword').value;

      const response = await fetch('php/login.php', {
        method: 'POST',
        body: JSON.stringify({ username, password }),
        headers: { 'Content-Type': 'application/json' }
      });

      const result = await response.json();
      if (result.message === "Login success") {
        window.location.href = "dashboard.html";
      } else {
        alert(result.message);
      }
    }

    async function register() {
      const username = document.getElementById("registerUsername").value;
      const email = document.getElementById("registerEmail").value;
      const password = document.getElementById("registerPassword").value;

      const response = await fetch('php/register.php', {
```

```

        method: 'POST',
        body: JSON.stringify({ username, email, password }),
        headers: { 'Content-Type': 'application/json' }
    });

    const result = await response.json();
    alert(result.message);
}

function showRegistration() {
    document.getElementById('registration').style.display = 'block';
    document.getElementById('login').style.display = 'none';
}

function showLogin() {
    document.getElementById('registration').style.display = 'none';
    document.getElementById('login').style.display = 'block';
}
</script>
</head>
<body class="bg-light">

    <div class="container mt-5">
        <div class="mb-3 d-flex justify-content-center">
            <button class="mr-2 btn btn-primary" onclick="showRegistration()">Register</button>
            <button class="btn btn-secondary" onclick="showLogin()">Login</button>
        </div>

        <div id="registration" class="p-4 shadow-sm card" style="display: none; max-width: 400px;
margin: auto;">
            <h2 class="text-center">Registration</h2>
            <div class="form-group">
                <label for="registerUsername">Username</label>
                <input type="text" class="form-control" id="registerUsername" placeholder="Username"
required>
            </div>
            <div class="form-group">
                <label for="registerEmail">Email Address</label>
                <input type="email" class="form-control" id="registerEmail" placeholder="Email
Address" required>
            </div>
            <div class="form-group">
                <label for="registerPassword">Password</label>
                <input type="password" class="form-control" id="registerPassword"
placeholder="Password" required>
            </div>
            <button class="btn btn-success btn-block" onclick="register()">Register</button>
        </div>

```

```

    <div id="login" class="p-4 shadow-sm card" style="display: block; max-width: 400px;
margin: auto;">
    <h2 class="text-center">Login</h2>
    <div class="form-group">
        <label for="loginUsername">Username</label>
        <input type="text" class="form-control" id="loginUsername" placeholder="Username"
required>
    </div>
    <div class="form-group">
        <label for="loginPassword">Password</label>
        <input type="password" class="form-control" id="loginPassword" placeholder="Password"
required>
    </div>
    <button class="btn btn-primary btn-block" onclick="login()">Login</button>
</div>
</div>

<!-- Bootstrap JS and dependencies (jQuery and Popper.js) -->
<script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
<script
src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"></script>
<script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</body>
</html>

```

dashboard.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard</title>
    <!-- Bootstrap CSS -->
    <link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">

    <script>
        async function logout() {
            const response = await fetch('php/logout.php', {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' }
            });
            const result = await response.json();
            if (result.message === "Logout success!") {

```

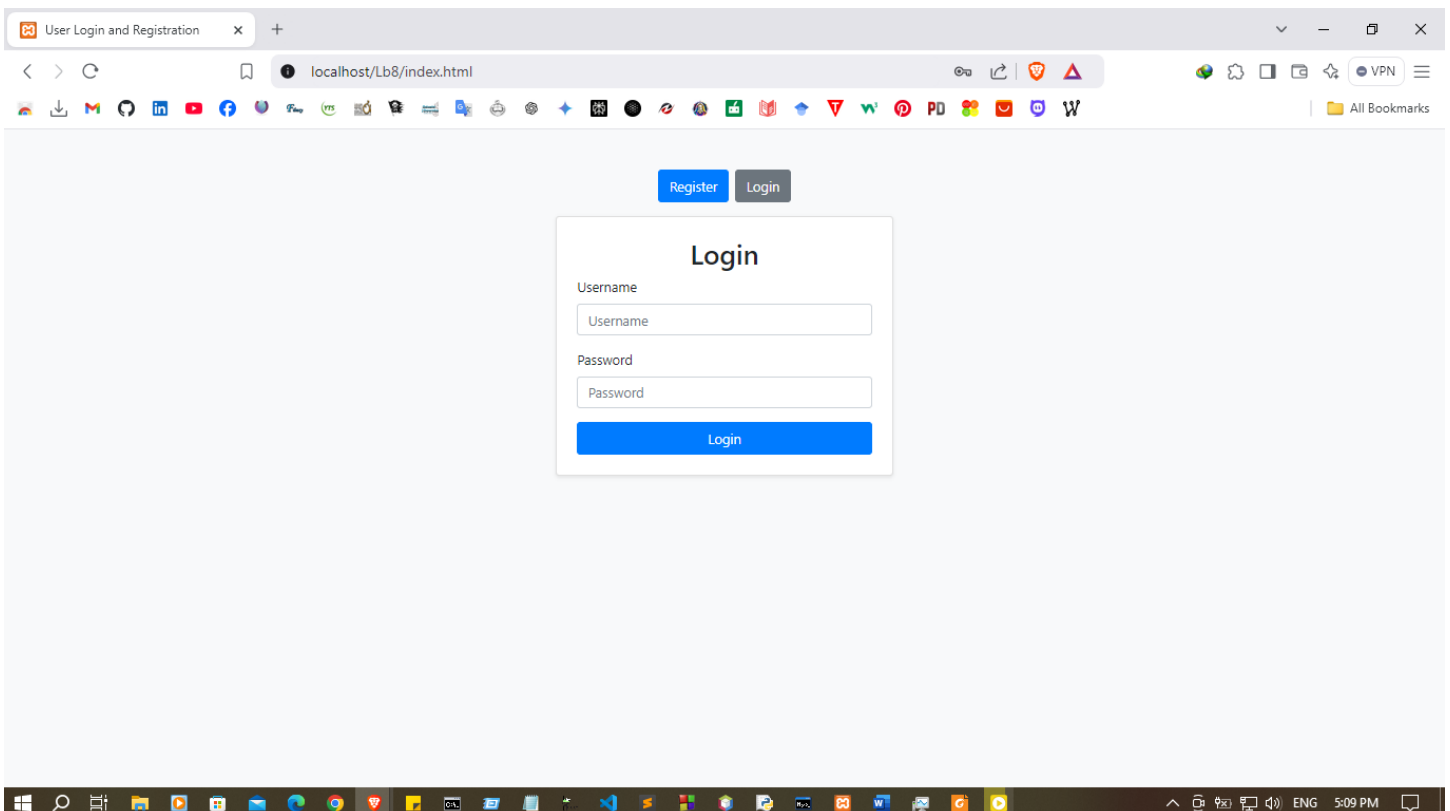
```

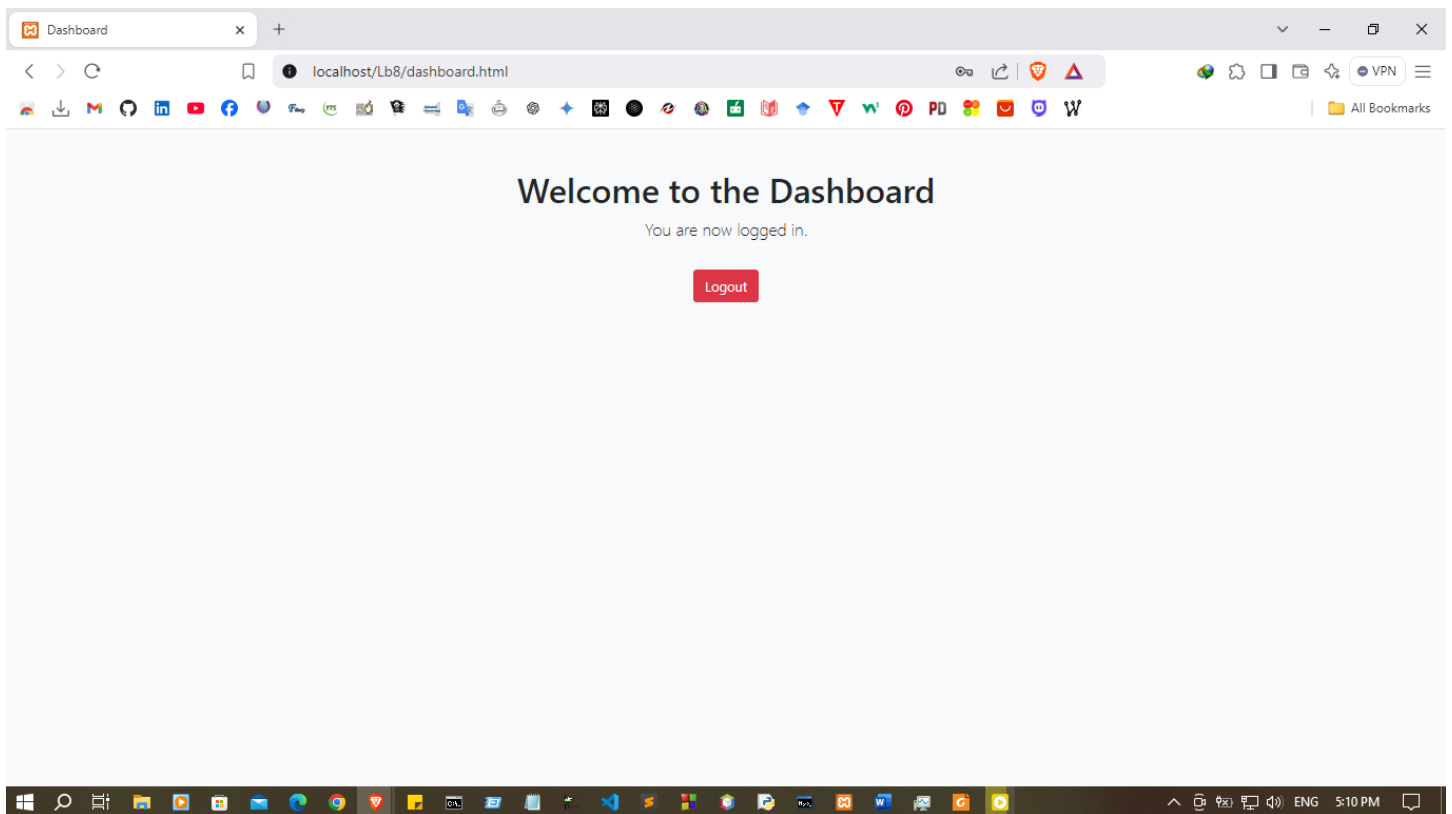
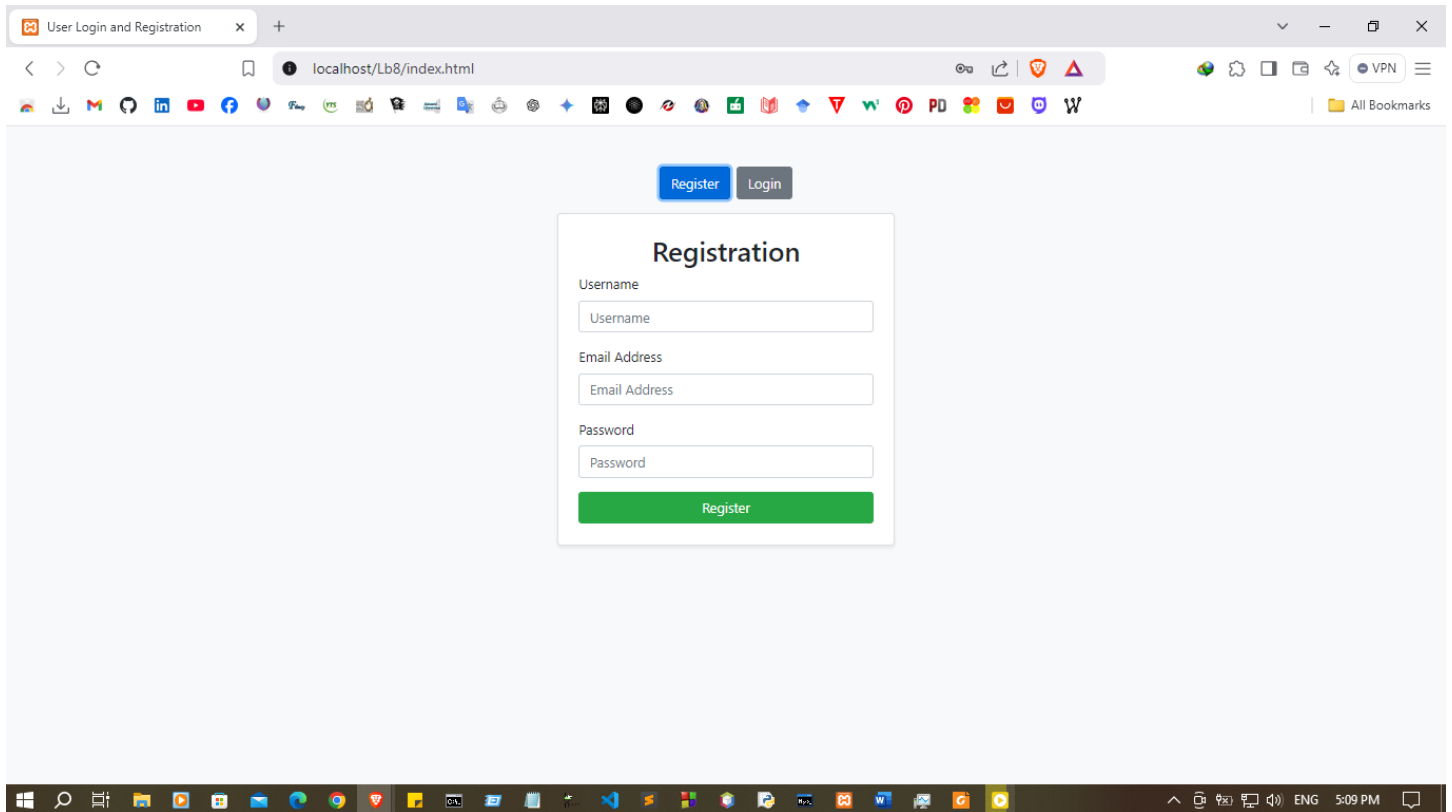
        window.location.href = "index.html";
    } else {
        alert("Error logging out. Please try again.");
    }
}
</script>
</head>
<body class="bg-light">
    <div class="container mt-5 text-center">
        <h1>Welcome to the Dashboard</h1>
        <p class="lead">You are now logged in.</p>
        <button class="mt-3 btn btn-danger" onclick="logout()">Logout</button>
    </div>

    <!-- Bootstrap JS and dependencies -->
    <script src="https://code.jquery.com/jquery-3.5.1.slim.min.js"></script>
    <script
src="https://cdn.jsdelivr.net/npm/popper.js@1.16.1/dist/umd/popper.min.js"></script>
    <script
src="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
</body>
</html>

```

OUTPUT





Discussion:

- In this lab practical session, I focused on developing a basic web application using PHP, MySQL, and JavaScript, with the main objective of understanding how these technologies work together to handle data and manage user sessions. First, I created a new MySQL database to store user information, which allowed me to practice setting up tables and defining data structures. Next, I implemented PHP functions to insert and retrieve data from the database, gaining hands-on experience with SQL queries for storing and accessing specific user data. This involved using PHP to communicate with MySQL, a crucial skill for backend development. I also worked on implementing a session management system, which allows users to log in and keep their session active as they navigate the web application. Finally, I integrated JavaScript for interactive elements, enhancing the user experience by providing dynamic, responsive feedback on the frontend. This session helped solidify my understanding of the interplay between these technologies in a real-world web application.